

ANSWERS

1.

Question	Answer	Marks
9(a)	$(T =) 2\pi r / v$ or $2\pi \times 1.1 \times 10^{11} / 3.5 \times 10^4$	C1
	2.0×10^7 (s)	A1
9(b)	it / speed does not have a direction	B1
9(c)(i)	it / velocity changes	M1
	(because) its <u>direction</u> changes	A1
9(c)(ii)	to change the velocity or to keep Venus in a circular orbit or to accelerate Venus towards the centre of the orbit / Sun	B1
9(c)(iii)	arrow towards Sun and on Venus	B1
9(c)(iv)	it / Venus is in the gravitational field or the gravitational attraction	M1
	of the Sun	A1
9(d)(i)	Mercury	B1
9(d)(ii)	time taken is less and orbital speed is greater / circumference of orbit is less / closer to the Sun	B1
	circumference of orbit is less and orbital speed is greater.	B1

2.

Question	Answer	Marks
10(a)(i)	distance travelled (in a vacuum) by light in one year	B1
10(a)(ii)	100 000	B1
10(b)(i)	supernova	B1
	nebula and neutron star	B1
10(b)(ii)	in a supernova	B1
10(c)	either red shift / increase in (observed) wavelength / reduction in frequency mentioned or galaxies / stars moving away (from the Earth) / separating	B1
	further away (the galaxy / star) the greater its speed / greater red shift / greater increase in wavelength	B1
	going backwards in time / at start) stars or galaxies were close together or high density / dense	B1
	cosmic microwave background mentioned or universe expands	B1
	remnant heat / radiation / left over radiation / radiation from early universe observed now	B1
	radiation has red shifted / longer wavelengths / smaller frequency / become cooler or CMBR is uniform / observed in all directions	B1